EduAssist AI: LLMOps Benchmarking & Prompt Lifecycle Report

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Assignment: Experiential Learning Assignment - 2 | **Tracking URI:** sqlite:///mlflow.db (http://127.0.0.1:5000)
Serving Backend: Ollama Local Runtime (llama3.2:1b) | **Experiment:** eduassist_llmops_assignment

1. Executive Summary & Objective

LLMOps extends conventional machine learning operations by treating prompts, prompt templates, system instructions, retrieval context, and generative outputs as version-controlled software assets. While traditional MLOps governs tabular features and deep neural net weights, LLMOps is responsible for autoregressive response latency, token consumption economics, guardrail integrity, and deterministic structural compliance. In this experiential project, we operationalize **EduAssist AI**, an automated academic QA system for undergraduate students. Using **MLflow 3.17.2** and local inference backend **Ollama (llama3.2:1b)**, we evaluate three prompt design paradigms across latency, token usage, cost economics, schema compliance, and lifecycle promotion.

Metric / Dimension	Prompt v1 (Basic)	Prompt v2 (Structured)	Prompt v3 (JSON Contract)
Prompt Architecture	Unconstrained baseline instruction	Academic persona & concise framing	Strict JSON schema constraint
Mean Response Latency	3.463s	1.999s (-42.3%)	**0.651s (-81.2% vs v1)**
Average Completion Tokens	448.6 tokens	363.2 tokens	**112.2 tokens (-75.0%)**
JSON Schema Pass Rate	0.0% (0 / 5 valid)	0.0% (0 / 5 valid)	**100.0% (5 / 5 valid)**
Marginal Token Cost	$0.00 (Self-hosted local)	$0.00 (Self-hosted local)	$0.00 (Self-hosted local)
Projected Cloud Cost / 10k Req	$2.76 / month	$2.28 / month	**$0.82 / month (-70.3%)**
MLflow Registry Promotion	Candidate v1	Candidate v2	**Promoted to @production**

2. Experimental Setup & Question Corpus

Testing was conducted across 5 foundational computer science questions: **(Q1)** Supervised vs Unsupervised learning differences; **(Q2)** Bias-variance tradeoff; **(Q3)** Overfitting causes and mitigation techniques; **(Q4)** Confusion matrix operational purpose; and **(Q5)** Gradient descent optimization mechanics. All evaluations were executed with deterministic greedy decoding (temperature = 0) using llama3.2:1b via the OpenAI SDK compatibility endpoint at http://localhost:11434/v1.

3. Task 1: API Calling & Latency Measurement

Question ID	Model Name	Start Timestamp	End Timestamp	Latency (s)	Response Status
Q1	llama3.2:1b	00:29:03.911	00:29:11.243	7.3325s	success (200 OK)
Q2	llama3.2:1b	00:29:11.243	00:29:13.975	2.7322s	success (200 OK)
Q3	llama3.2:1b	00:29:13.976	00:29:16.536	2.5604s	success (200 OK)
Q4	llama3.2:1b	00:29:16.536	00:29:18.380	1.8446s	success (200 OK)
Q5	llama3.2:1b	00:29:18.380	00:29:21.223	2.8429s	success (200 OK)

Prompt v1 Latency Statistics: Mean: **3.4625s** | Median: **2.7322s** | Min: **1.8446s** (Q4) | Max: **7.3325s** (Q1) | Std Dev: **2.1980s**.
Analysis of Latency Variance: Request Q1 exhibited the highest latency (7.3325s), while Q4 completed fastest (1.8446s). Because autoregressive LLM decoding operates token-by-token, latency scales proportionally with generated completion length (O(N_out)). Furthermore, Q1 experienced engine initialization and KV-cache cold-start warmup delay. Subsequent requests (Q2-Q5) maintained steady latencies around 2.5-2.8s.

4. Task 2: Token Usage, Cost Economics & Optimization

Question ID	Input Tokens	Output Tokens	Total Tokens	Input Cost ($)	Output Cost ($)	Cost / Req ($)
Q1	46	422	468	$0.000000	$0.000000	$0.000000
Q2	42	497	539	$0.000000	$0.000000	$0.000000
Q3	46	467	513	$0.000000	$0.000000	$0.000000
Q4	43	337	380	$0.000000	$0.000000	$0.000000
Q5	40	520	560	$0.000000	$0.000000	$0.000000
Average	**43.4**	**448.6**	**492.0**	**$0.000000**	**$0.000000**	**$0.000000**

Projected Monthly Cost (10,000 queries): In the local Ollama runtime, marginal billing is $0.00/month. Projecting onto standard commercial cloud pricing (e.g., gpt-4o-mini at $0.15/1M input and $0.60/1M output tokens): Prompt v1 incurs $0.000276 per request, resulting in **$2.76 / 10k requests**. **Strategic Cost-Optimization Recommendation:** Output tokens are billed at 4x the rate of input tokens across commercial LLM providers. Unconstrained prompt templates (v1) encourage verbosity and redundant pleasantries (448.6 output tokens). By enforcing rigid JSON schema constraints (Prompt v3), generated completion tokens are capped to 112.2 tokens, reducing recurring operating expenditure by **70.3% ($0.82 / month)** while simultaneously eliminating downstream JSON parsing exceptions.

5. Task 3: Prompt Versioning & MLflow Evaluation

Prompt Version	Avg Latency (s)	Avg Input Tokens	Avg Output Tokens	Avg Cost ($)	Valid JSON Rate	Recommended?
v1 (Basic)	3.463s	43.4	448.6	$0.000000	0.0% (0/5)	No
v2 (Structured)	1.999s	65.4	363.2	$0.000000	0.0% (0/5)	No
v3 (JSON Contract)	**0.651s**	99.4	**112.2**	$0.000000	**100.0% (5/5)**	**Yes (@production)**

6. Production Promotion Justification (>150 Words)

Prompt version **v3** is definitively recommended and promoted to the **@production** alias in the MLflow Prompt Registry for the EduAssist platform. In enterprise LLMOps systems, automated downstream consumers and web frontends require strict, deterministic contracts rather than arbitrary prose. Both Prompt v1 and Prompt v2 exhibited an unacceptable **0.0% valid JSON rate**, generating markdown formatting and narrative paragraphs that break API response parsers. In stark contrast, Prompt v3 delivered a flawless **100.0% valid JSON pass rate (1.00)**, strictly fulfilling the target contract (answer, confidence, key_points). Beyond schema reliability, Prompt v3 demonstrates dramatic operational superiority: average inference latency dropped from **3.463 seconds (v1) to 0.651 seconds (v3)**, delivering an extraordinary **81.2% speedup in user-perceived responsiveness**. This acceleration directly results from a **75.0% reduction in generated tokens** (from 448.6 down to 112.2 completion tokens). Because autoregressive LLM decoding computes tokens sequentially, minimizing completion volume reduces GPU memory bandwidth pressure and network overhead. On commercial cloud tiers, v3 cuts projected recurring cost by **70.3% ($0.82 vs $2.76 per 10k calls)**. With guaranteed structural integrity, ultra-low latency, and optimal cost efficiency, Prompt v3 is the unequivocal production choice.